

# Week 7 - Sensitivity analysis report

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\*\*Report date\*\*: 20260619 | \*\*Pipeline version\*\*: v1.0

### 1. Tool Configuration

- Model artifact: week6\_approach2\_neuralnetwork\_v1.0.joblib
- Reference date: 2024-12-31
- Input feature count: 23
- Baseline model price: 24.423695
- Closed-form reference: 33.429823
- Monte Carlo reference: 33.602667

### 2. Extreme Scenario Tests

| scenario               | model_price | delta     | delta_pct | closed_form_quote | mc_quote  |
|------------------------|-------------|-----------|-----------|-------------------|-----------|
| baseline               | 24.423695   | 0         | 0         | 33.429823         | 33.602667 |
| volatility_spike_50pct | 25.944422   | 1.520727  | 0.062264  | 50.218133         | 49.108253 |
| rate_hike_2pct         | 24.383972   | -0.039724 | -0.001626 | 33.48704          | 33.653419 |
| sentiment_shock        | 28.007684   | 3.583989  | 0.146742  | 33.429823         | 33.602667 |
| combined_stress        | 28.184373   | 3.760677  | 0.153977  | 50.116995         | 49.170447 |

### 3. Sensitivity Grid

| feature       | value    | model_price | delta     | delta_pct |
|---------------|----------|-------------|-----------|-----------|
| vix           | 12.145   | 23.137409   | -1.286287 | -0.052666 |
| vix           | 15.615   | 23.886675   | -0.537021 | -0.021988 |
| vix           | 17.35    | 24.423695   | 0         | 0         |
| vix           | 19.085   | 25.074049   | 0.650353  | 0.026628  |
| vix           | 26.025   | 29.027987   | 4.604292  | 0.188517  |
| sentiment_7d  | 0        | 24.825158   | 0.401462  | 0.016437  |
| sentiment_7d  | 0.005333 | 24.802069   | 0.378373  | 0.015492  |
| sentiment_7d  | 0.105333 | 24.423695   | 0         | 0         |
| sentiment_7d  | 0.205333 | 24.950535   | 0.526839  | 0.021571  |
| sentiment_7d  | 0.355333 | 26.611322   | 2.187626  | 0.08957   |
| sentiment_20d | 0        | 28.578059   | 4.154364  | 0.170096  |
| sentiment_20d | 0.04583  | 27.292497   | 2.868801  | 0.11746   |
| sentiment_20d | 0.14583  | 24.423695   | 0         | 0         |

| feature               | value   | model_price | delta     | delta_pct |
|-----------------------|---------|-------------|-----------|-----------|
| sentiment_20d         | 0.24583 | 23.190763   | -1.232933 | -0.050481 |
| sentiment_20d         | 0.34583 | 24.649922   | 0.226227  | 0.009263  |
| r                     | 0.0358  | 25.324733   | 0.901038  | 0.036892  |
| r                     | 0.0458  | 24.423695   | 0         | 0         |
| r                     | 0.0558  | 24.398439   | -0.025256 | -0.001034 |
| r                     | 0.0658  | 24.383972   | -0.039724 | -0.001626 |
| q                     | 0.01419 | 26.400603   | 1.976908  | 0.080942  |
| q                     | 0.01919 | 24.423695   | 0         | 0         |
| q                     | 0.02419 | 23.034311   | -1.389385 | -0.056887 |
| T2                    | 0.3     | 22.393345   | -2.03035  | -0.08313  |
| T2                    | 0.4     | 23.36571    | -1.057985 | -0.043318 |
| T2                    | 0.5     | 24.423695   | 0         | 0         |
| _7 more rows omitted_ |         |             |           |           |

#### 4. SHAP Impact Summary

| feature          | mean_abs_shap |
|------------------|---------------|
| vix_ma_ratio     | 1.316436      |
| vix              | 0.869999      |
| vix_jpm_corr_20d | 0.56082       |
| vol_ratio_20_60  | 0.346856      |
| sentiment_7d     | 0.282197      |
| q                | 0.276816      |
| vix_change_5d    | 0.26088       |
| sentiment_20d    | 0.244439      |
| vol_ratio_5_20   | 0.176465      |
| return_20d       | 0.126048      |

#### 5. Observations

- The pricing tool now exposes a direct quote, a closed-form reference, and a Monte Carlo reference for the same contract.
- Scenario tests include a 50% volatility spike, a 2% rate hike, a sentiment shock, and a combined stress case.
- Sensitivity grids are centered on the latest usable market row so the tool can be refreshed with new daily data.